

SECTION 1: Aquatic Ecology & Water Systems

1. During thermal stratification in summer, which process causes oxygen depletion in the hypolimnion of a lake?
 - A) Increased photosynthesis by algae
 - B) Decomposition of organic matter without mixing from surface waters
 - C) Excessive evaporation at the thermocline
 - D) Rising water temperature increasing oxygen solubility

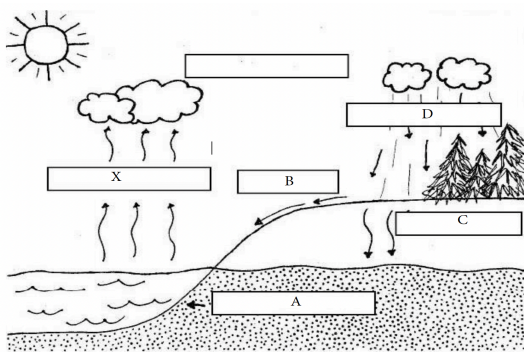
2. In aquatic food webs, organisms that produce their own food through photosynthesis are called:
 - A) Primary consumers
 - B) Decomposers
 - C) Primary producers
 - D) Tertiary consumers

3. A farmer's field borders a stream that is experiencing frequent algal blooms and low dissolved oxygen. Which is the most likely source of pollution causing eutrophication?
 - A) Thermal pollution from a nearby power plant
 - B) Sediment runoff from construction
 - C) Acid rain from industrial emissions
 - D) Agricultural fertilizer runoff containing nitrogen and phosphorus

4. Which type of community interaction occurs when zebra mussels outcompete native mussels for food and space?
 - A) Predation
 - B) Mutualism
 - C) Competition
 - D) Parasitism

5. A town is experiencing increased flooding and poor water quality downstream. Which watershed management practice would most effectively address both problems?
 - A) Building a dam to store water
 - B) Installing riparian buffer zones with native vegetation
 - C) Dredging the stream channel to make it deeper
 - D) Paving areas near the stream to prevent erosion

Questions 6-7 refer to the Water Cycle Diagram



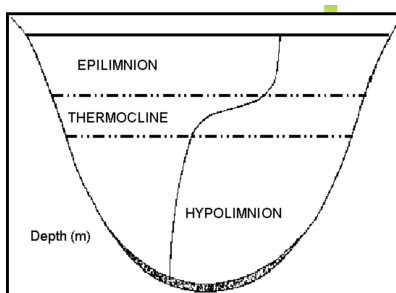
6. In the water cycle diagram, which process is represented by box X showing water moving from the ocean surface into the atmosphere?

- ☐ A) Condensation
- ☐ B) Evaporation
- ☐ C) Transpiration
- ☐ D) Precipitation

7. Which letter on the diagram shows water infiltrating into groundwater?

- ☐ A) Point A
- ☐ B) Point B
- ☐ C) Point C
- ☐ D) Point D

Question 8 refer to the Lake Zones Diagram

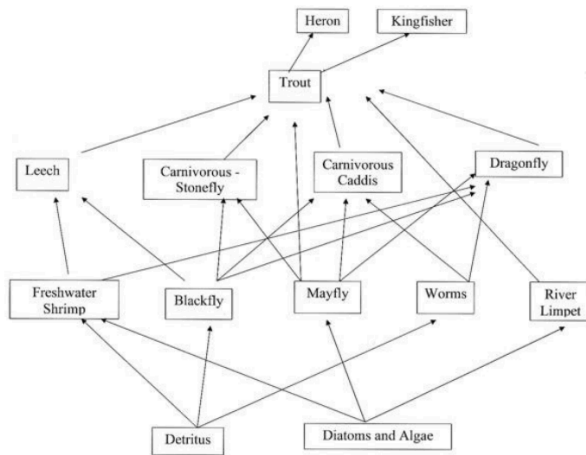


8. Rooted aquatic plants would most likely be found in which zone?

- ☐ A) epilimnion
- ☐ B) thermocline
- ☐ C) hypolimnion
- ☐ D) Abyssal zone

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Question 9-10 refers to the Aquatic Food Web Diagram



9. If a pollutant enters the food web at the algae level, which organism would likely have the highest concentration due to biomagnification?

- A. Freshwater shrimp
 - B. trout
 - C. kingfisher
 - D. All would have equal amounts
10. Which of these pairs are competing for the same food source?
- A) Leech, limpet
 - B) Stonefly, dragonfly
 - C) Trout, kingfisher
 - D) Mayfly, worms

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Use this WORD BANK for questions 11-20: secondary, invasive, usable, littoral, eutrophication, coagulation, epilimnion, potable, decomposers, sedimentation, primary, native, profundal, bioaccumulation, filtration, biomagnification, primary

Note: Not all words will be used. Words may be used more than once.

11. The _____ zone of a lake is the shallow area where sunlight penetrates to the bottom, allowing rooted plants to grow.
 12. In wastewater treatment, the _____ treatment stage uses bacteria to break down organic matter.
 13. The nitrogen cycle converts atmospheric nitrogen into _____ forms that plants can absorb.
 14. When a species is introduced to a new ecosystem and causes harm, it is called an _____ species.
 15. The process where excess nutrients lead to algal blooms and oxygen depletion is called _____.
 16. During water treatment, _____ is the process where chemicals like alum cause particles to clump together.
 17. The buildup of toxins within a single organism over its lifetime is called _____.
 18. Water that is safe for human consumption is called _____ water.
 19. Bacteria and fungi that break down dead organic matter in aquatic ecosystems are called _____.
 20. The _____ of particles is an important step in both natural water systems and water treatment plants.
- 21.** Explain two specific ways excessive sedimentation pollution impacts aquatic ecosystems.
- 22.** As water temperature increases, dissolved oxygen levels (increase/decrease).

Explain why

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SECTION 2: Macroinvertebrate & Aquatic Species Identification (30 points)

Identify which pollution tolerance class (1, 2, 3, 4, or 5) each organism belongs to:

23. Stonefly: _____

24. Dragonfly: _____

25. Blood midge: _____

26. Water strider: _____

27. Dobsonfly: _____

28) If a stream sample contains mostly mayflies, stoneflies, and caddisflies, what does this indicate about water quality?

- A) Poor water quality - these are pollution tolerant organisms
- B) Excellent water quality - these are pollution sensitive organisms
- C) Moderate water quality - these organisms tolerate some pollution
- D) Cannot determine water quality from these organisms

29) Which pollution tolerance class would you expect to find in the greatest numbers in a stream receiving untreated sewage discharge?

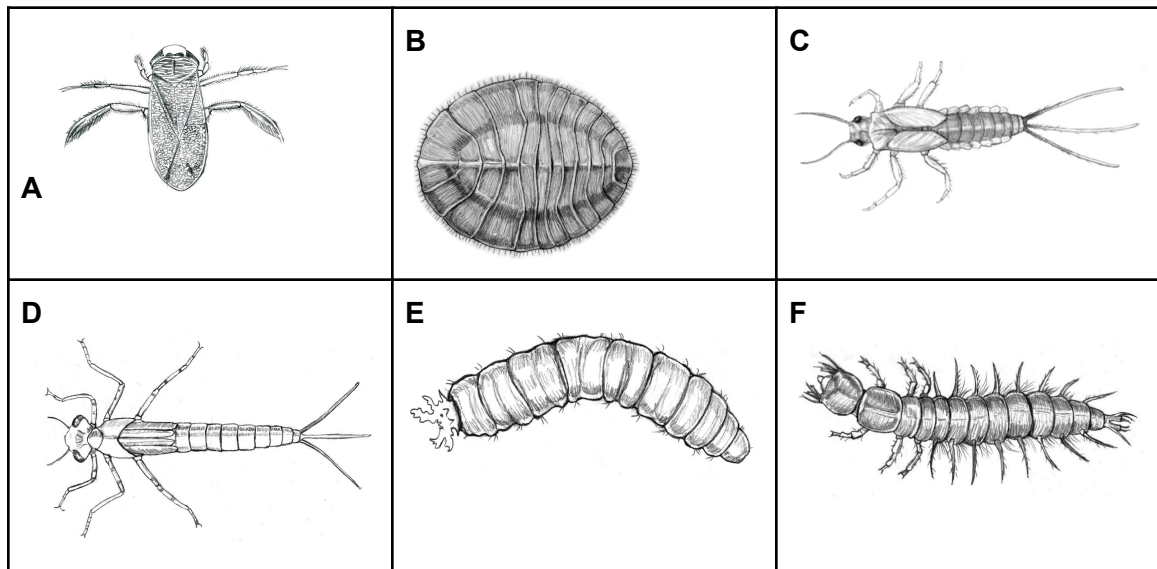
- A) Class 1
- B) Class 2
- C) Class 4
- D) Class 5

30. Identify two specific ways purple loosestrife impacts native wetland ecosystems.

31. Identify two specific ways zebra mussels impact aquatic ecosystems.

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IDENTIFY THE FOLLOWING ORGANISMS with their common names



32. Normal pH range for most healthy aquatic ecosystems: _____

- a. 3.5-5.5
- b. 6.5-8.5
- c. 9.5-11.5
- d. 12.0-14.0

33. High turbidity in water primarily blocks:

- a. Sound waves
- b. Wind
- c. Sunlight
- d. Evaporation

34. Dissolved oxygen levels below _____ mg/L are generally considered stressful for most fish:

- | | |
|-------|-------|
| a. 10 | c. 5 |
| b. 8 | d. 15 |

35. Salinity is a measure of:

- a. Water temperature
- b. Oxygen content
- c. Acidity
- d. Dissolved salts

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36. A Secchi disk is lowered into a lake and disappears from view at 2 meters depth. What does this measurement indicate?

- a. The lake has high dissolved oxygen at 2 meters
- b. The lake has low turbidity and excellent water clarity
- c. The lake has high turbidity and poor water clarity
- d. The lake temperature changes at 2 meters

37. Fecal coliform bacteria in water indicate:

- a. High oxygen levels
- b. Low Temperature
- c. Excessive plant growth
- d. Sewage or animal waste contamination

38. High BOD indicates:

- A) Clean, unpolluted water
- B) High levels of organic matter requiring oxygen to decompose
- C) Excessive salinity
- D) Low turbidity

39. The presence of phosphates in high concentrations typically leads to:

- A) Decreased plant growth
- B) Algal blooms
- C) Increased water clarity
- D) Higher dissolved oxygen

Data Interpretation (22 points total)

Scenario 1: A research team collected data from three stream sites over one month.

Site	pH	Dissolved O ₂ (mg/L)	Turbidity (NTU)	Nitrates (mg/L)	Phosphate (mg/L)	Temp (°C)
A	7.2	9.5	8	0.5	0.02	15
B	6.8	4.2	45	8.5	1.8	22
C	7.8	7.8	12	1.2	0.15	16

40. Which site shows the best overall water quality?

- A) Site A
- B) Site B
- C) Site C

41. What type of pollution is most likely affecting Site B's nitrates and phosphates?

- A) Thermal pollution from power plants
- B) Agricultural runoff or sewage
- C) Acid rain
- D) Heavy metal contamination

42. Why are the high nitrate and phosphate levels at Site B concerning? (2 points)

- A) They make the water too acidic for fish
- B) They cause eutrophication and algal blooms leading to oxygen depletion
- C) They increase water temperature
- D) They make the water unsafe to drink but don't harm aquatic life

Scenario 2: A lake is tested monthly during the summer.

Month	Temp (°C)	DO (mg/L)	Phosphate (mg/L)	Turbidity (NTU)	pH
June	18	8.5	0.08	15	7.4
July	24	6.2	0.35	32	8.1
Aug	26	4.8	0.52	58	8.6

43. What is happening to water quality from June to August? (2 points)
- A) Improving - all parameters are moving toward optimal ranges
 - B) Declining - all parameters are moving toward sub-optimal ranges
 - C) Staying stable with normal seasonal changes
 - D) Cannot determine from this data
44. What ecological process is responsible for these changes? (2 points)
- A) Eutrophication
 - B) Thermal stratification
 - C) Denitrification
 - D) Biomagnification
45. Based on the August data, would you expect to find mostly Class 1 or Class 4 macroinvertebrates?
- A) Class 1 - pollution sensitive organisms thrive in these conditions
 - B) Class 4 - pollution tolerant organisms because of low DO and poor water quality
 - C) Class 1 - pollution sensitive organisms need high oxygen levels like those shown
 - D) Class 4 - pollution tolerant organisms prefer warm temperatures
46. If this lake were sampled in September after a cold front dropped temperatures to 16°C, what would likely happen to dissolved oxygen levels? (2 points)
- A) DO would increase
 - B) DO would decrease
 - C) DO would stay the same
 - D) DO would become zero
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